

OSTROV ANDREYA, Russian Federation

WMO: 213010

Lat: 76.779N

Lon: 110.787E

Elev: 20

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.5	-43.8	-51.0	0.3	-47.1	-47.7	0.3	-43.7	53.4	-7.3	47.3	-7.9	7.2	210	1.221

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	54.6	51.2	49.0	46.3	45.0	42.9	51.4	54.5	46.5	48.7	43.1	44.8	10.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.1	51.7	53.9	44.6	43.5	48.3	41.4	38.4	44.5	21.1	54.5	18.5	48.9	16.7	44.9	70.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 37.9	32.1	28.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)			WB	-50.2	59.0	4.2	5.8	-53.2	63.2	-55.7	66.5	-58.0	69.8	-61.1	74.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-24.0	-24.1	-19.9	-9.2	13.2	29.6	37.6	36.2	27.0	4.5	-14.6	-18.8
	DBStd	25.76	14.21	12.83	13.10	11.46	8.20	5.53	5.71	5.84	7.93	13.68	11.13	12.13
	HDD50	17059	2293	2076	2167	1777	1141	612	391	432	689	1410	1939	2132
	HDD65	22525	2758	2496	2632	2227	1606	1062	850	894	1139	1875	2389	2597
	CDD50	9	0	0	0	0	0	0	6	3	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	12.7	12.6	12.6	13.1	13.3	11.6	12.7	11.9	12.7	12.6	13.4	12.3	13.9
	PrecAvg	10.6	0.6	0.5	0.5	0.6	0.9	0.8	1.2	1.5	1.5	1.3	0.8	0.6
(15) Precipitation	PrecMax	16.2	2.0	1.3	0.9	1.8	1.7	1.6	2.2	3.0	3.3	2.5	1.5	1.6
	PrecMin	5.8	0.0	0.0	0.0	0.0	0.1	0.0	0.2	0.4	0.5	0.5	0.3	0.1
	PrecStd	2.8	0.4	0.4	0.3	0.4	0.4	0.4	0.5	0.7	0.7	0.4	0.4	0.4
	PrecStd	2.8	0.4	0.4	0.3	0.4	0.4	0.4	0.5	0.7	0.7	0.4	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	18.2	13.1	19.9	27.2	34.6	48.9	63.7	63.7	49.8	33.4	20.5	16.8	
	0.4% MCWB	17.8	12.5	19.1	26.4	33.6	46.3	59.1	58.0	47.2	31.3	19.7	15.6	
	2% DB	9.1	5.5	12.8	18.3	30.9	41.6	56.3	54.6	43.9	30.4	11.5	8.6	
	2% MCWB	8.4	4.8	11.8	17.2	29.5	39.3	52.6	51.2	41.9	29.3	10.9	7.8	
	5% DB	1.9	1.7	5.6	13.4	27.3	37.2	50.1	48.9	38.8	27.2	5.6	3.8	
	5% MCWB	1.4	1.2	5.0	12.6	26.0	35.3	47.4	46.2	37.1	26.2	5.0	3.3	
	10% DB	-2.6	-4.4	-0.5	8.7	24.8	35.0	45.5	45.1	36.3	23.9	1.4	-1.1	
	10% MCWB	-2.9	-4.7	-1.0	8.1	23.7	33.5	43.4	43.2	35.1	22.9	0.8	-1.5	
	0.4% WB	17.0	12.7	18.8	26.4	33.4	46.4	60.3	58.4	46.9	31.7	20.1	15.7	
	0.4% MCDB	18.0	13.2	19.4	27.6	34.4	48.9	63.6	63.3	49.3	32.9	20.7	16.4	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	8.5	5.1	11.5	17.5	29.5	39.1	52.6	51.4	41.9	29.3	10.8	8.0	
	2% MCDB	9.2	5.8	12.3	18.4	30.8	41.5	56.0	54.4	44.1	30.4	11.2	8.4	
	5% WB	1.6	1.2	5.0	12.8	26.1	35.5	47.6	46.5	37.2	26.2	5.1	3.3	
	5% MCDB	1.8	1.8	5.5	13.5	27.3	36.9	50.1	48.6	38.4	27.1	5.5	3.7	
	10% WB	-3.0	-4.8	-1.1	8.1	23.7	33.8	43.6	43.3	35.1	22.9	1.0	-1.4	
	10% MCDB	-2.6	-4.4	-0.6	8.7	24.7	34.7	45.5	45.1	36.1	23.9	1.5	-1.0	
(35) Mean Daily Temperature Range	MDBR	11.1	11.8	12.8	14.0	9.8	5.2	7.7	7.4	6.9	11.2	11.8	11.0	
	5% DB	13.9	19.2	17.7	15.3	10.5	8.8	18.4	17.1	9.6	10.8	17.9	14.1	
	MCWBR	13.4	18.9	17.7	15.1	9.9	6.6	15.6	14.0	7.9	10.6	17.6	13.8	
	5% WB	14.0	18.6	18.8	15.2	10.4	8.1	18.5	16.8	8.8	11.4	17.9	14.0	
	MCWBR	13.5	18.4	18.4	15.0	9.7	6.5	15.8	14.0	7.5	11.2	17.7	13.7	
	MCWBR	13.5	18.4	18.4	15.0	9.7	6.5	15.8	14.0	7.5	11.2	17.7	13.7	
(40) Clear-Sky Solar Irradiance	taub	N/A	0.171	0.258	0.296	0.321	0.318	0.354	0.334	0.273	0.186	N/A	N/A	
	taud	N/A	1.874	1.985	2.047	2.075	2.307	2.352	2.417	2.342	1.949	N/A	N/A	
	Ebn at Noon	0	41	181	238	254	265	246	232	193	28	0	0	
	Edh at Noon	0	3	19	31	36	30	28	23	14	2	0	0	
(44) All-Sky Solar Radiation	RadAvg	0	15	326	1071	1613	1626	1311	803	306	47	0	0	
	RadStd	0	2	30	88	141	193	108	84	33	6	0	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page